

Spring 2021, MATH 8120 Real Analysis II, Midterm 1

PRINT NAME: _____ **ID#:** _____

- Show your work. Answers with no or incorrect steps will not receive partial credit.
- The exam is self-explanatory. The instructor will not interpret the questions.

1. (4 points) Let $(X, \|\cdot\|)$ be a Banach space and X_0 a linear subspace of X . Suppose there exists $c \in (0, 1)$ such that

$$c\|y\| \geq \inf_{x \in X_0} \|y - x\|$$

for all $y \in X$. Show that X_0 is dense in X .

2. (4 points) Let (X, d) be a complete metric space and E is a closed subset of X . Show that E is sequentially precompact if and only if for any $\epsilon > 0$ there exists a sequentially precompact ϵ -net of E .

3. (4 points) Let (X, d) be a complete metric space and $T : X \rightarrow X$ a mapping. Suppose $T^n : X \rightarrow X$ is contractive. Show that T has a unique fixed point in X .

4. (4 points) Let $(X, \|\cdot\|)$ be a Banach space.

(a) Show that every totally bounded subset E of X is bounded.

(b) Let $l^2 := \{x = (x_1, \dots) : x_i \in \mathbb{R}\}$ with norm $\|x\| = (\sum_{i=1}^{\infty} |x_i|^2)^{1/2}$ for any $x \in l^2$. Show that the set $E := \{e_k : k \in \mathbb{N}\}$, where

$$e_k = (\underbrace{0, \dots, 0}_{k \text{ terms}}, 1, 0, \dots),$$

is bounded but not totally bounded in $(l^2, \|\cdot\|)$.

5. (4 points) Let (M, d) be a compact metric space and $C(M) := \{f : M \rightarrow \mathbb{R} : f \text{ is continuous}\}$ with norm $\|f\| := \max_{x \in M} |f(x)|$. Let $E \subset C(M)$. Suppose there exist $c, C > 0$, $\alpha \in (0, 1]$, and $\hat{x} \in M$, such that

$$|f(\hat{x})| \leq C \quad \text{and} \quad |f(x_1) - f(x_2)| \leq c d(x_1, x_2)^\alpha$$

for all $x_1, x_2 \in M$ and $f \in E$. Show that E is sequentially precompact.

All problems above are also shown in this page.

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$$c\|y\| \geq \inf_{x \in X_0} \|y - x\|$$

for all $y \in X$. Show that X_0 is dense in X .

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